



Artificial Intelligence Software, HW 8

Gradio



Center for
Cognitive
Modeling



Goal of the Assignment

Short description: Build interactive Gradio web applications covering REST API integration, UI components, and ML model deployment.

1. Get familiar with the Gradio framework for building interactive web demos.
2. Integrate external REST APIs with various input/output components.
3. Use advanced Gradio layout components: tabs, gallery, dataframe, JSON.
4. Build a Gradio interface for the YOLO instance segmentation model from HW 7.



Task 1

Short description: Build a joke application and extend it with automatic Russian translation.

Artifacts: Code (.py files) and PDF report with screenshots for each subtask.

Requirements

1. Implement a Gradio app that fetches and displays a random joke on button press.

Use API: https://official-joke-api.appspot.com/random_joke

2. Extend the app: add a button to translate the joke into Russian.

Suggested model: facebook/nllb-200-distilled-600M



Task 2

Short description: Build a weather application and an image gallery application.

Artifacts: Code (.py files) and PDF report with screenshots for each subtask.

Requirements

1. Implement a Gradio app that returns temperature and wind speed by user-provided coordinates. The user selects the temperature unit: Celsius or Fahrenheit. Use API: <https://open-meteo.com/>



Task 3

Short description: Implement a full CRUD GUI for a REST API using Gradio tabs.

Artifacts: Code (.py files) and PDF report with screenshots for each operation.

Requirements Implement a Gradio app providing GUI for /posts at <https://jsonplaceholder.typicode.com/>. Each CRUD operation is placed on a separate tab; HTTP response body is shown in gr.JSON.

- **Read N:** fetch first N posts → gr.DataFrame with fields *userId*, *id*, *title*, *body*
- **Read one:** fetch post by ID → fields: *userId*, *title*, *body*
- **Create:** create new post → fields: *userId*, *title*, *body*
- **Update:** update post by ID → fields: *userId*, *title*, *body*
- **Delete:** delete post by ID

Reference: <https://huggingface.co/learn/llm-course/en/chapter9/7>



Task 4

Short description: Build a Gradio interface for the YOLO instance segmentation model deployed via Triton Inference Server (with FastAPI) from HW 7.

Artifacts: Code (.py files) and PDF report with screenshots.

Requirements

- Reuse the full deployment stack from HW 7 (Triton + FastAPI service).
- The user uploads an image in the Gradio UI; the app sends it to the **FastAPI** endpoint from HW 7 and displays the returned annotated image with segmentation masks and class labels overlaid.
- Inference must go through the HW 7 service stack, not directly through the YOLO model.

Deserve "A" grade!

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